



Model Context Protocol: A New Genie for Crystallography



Slides: <https://github.com/Tae-Sung/Public>

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Teaching crystallographic computing

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Abstract No. 432

The Barrier: Crystallographic Computing is Hard to Teach

Structure determination is a chain of command-line tools, each with its own syntax, log-file conventions, and tacit decision points.

Command-line syntax

Beginners spend effort on flags and file paths rather than on diffraction physics.

Opaque log files

Critical diagnostics are buried in verbose text output with no guided interpretation.

Tacit decisions

Resolution cut-offs and refinement strategy rely on experience that is rarely written down.

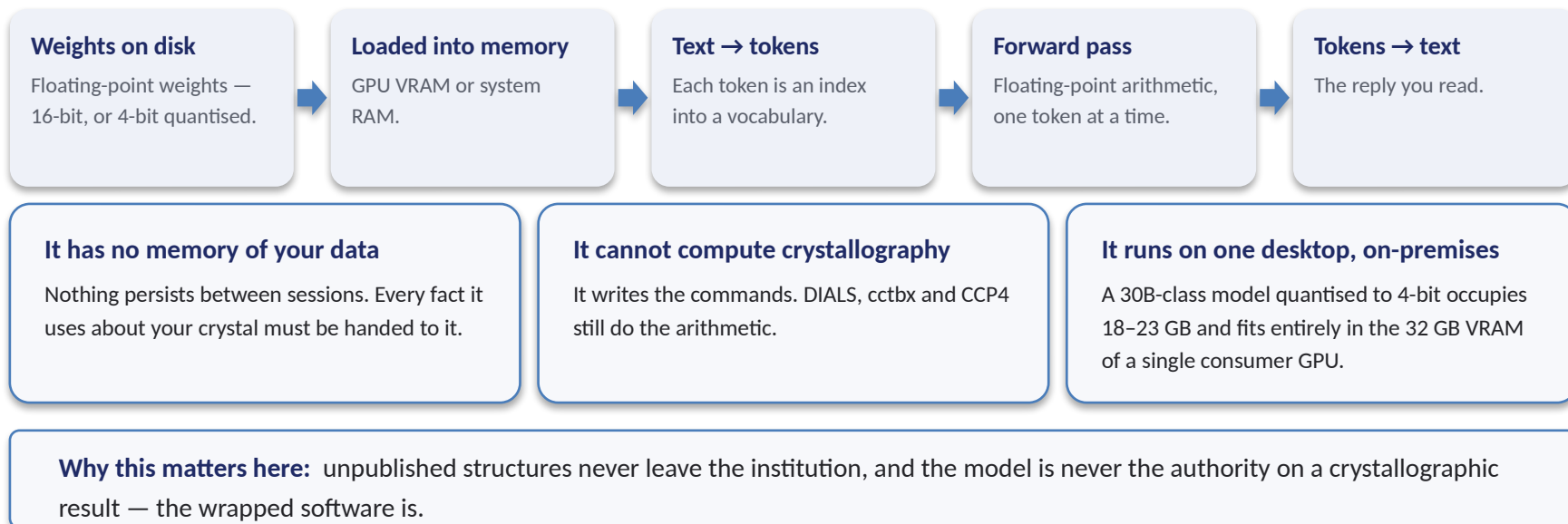
No ground truth

With experimental data the correct answer is unknown, so a mistake looks like a result.

Consequence: the conceptual content of crystallography is gated behind operational friction.

Bigger Picture I — What an LLM Actually Does on Your Machine

Before trusting a model with a workflow, it helps to be concrete about what it is and what it is not.



Bigger Picture II — Where MCP^[1] Sits in the Agentic Landscape

The interoperability layer settled during 2025–2026. Two protocols now cover distinct problems, and crystallography only needs one of them today.

Applications

Chat clients, coding agents, notebooks, lab dashboards

Agent runtime

Planning, tool selection, retry and error handling

Interoperability protocols

MCP — agent ↔ tool A2A — agent ↔ agent

Domain software

DIALS · cctbx / nanoBragg · CCP4 · instrument control

MCP Anthropic, Nov 2024 · Linux Foundation AAIF since Dec 2025 · 18,000+ community servers. The de facto agent–tool standard.

A2A Google, Apr 2025 · Linux Foundation · v1.0 Apr 2026 · 150+ organisations. IBM's ACP merged into it in Aug 2025.

MCP as a Uniform Interface Layer

MCP^[1] wraps crystallographic software as uniform services. What sits on top of that layer is interchangeable — today a deterministic interface, next a conversational one.

RS-Genie Studio — guided interface (shipped)

LLM chat layer — RS-Genie (in development)

Model Context Protocol — uniform tool description, invocation, and result schema

DIALS^[2]

Indexing, integration, scaling
with cctbx^[3] and nanoBragg^[4]

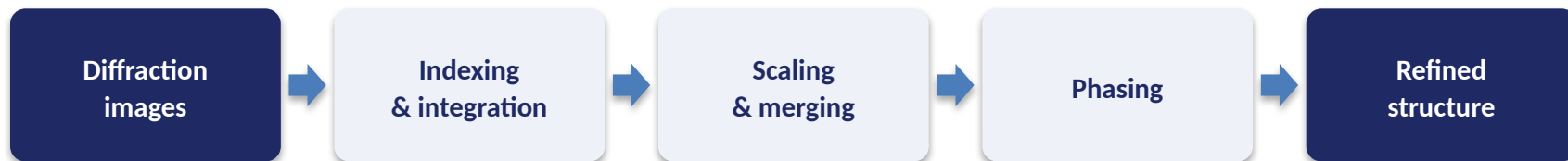
CCP4^[5]

Phasing, density modification,
model building, refinement

Current scope: Phenix is not included in this implementation. The wrapped set is DIALS (with cctbx and nanoBragg) and CCP4.

End-to-End Agentic Workflow

The pipeline runs the full path from diffraction images to a refined model, and surfaces the diagnostics behind every step instead of burying them in logs.



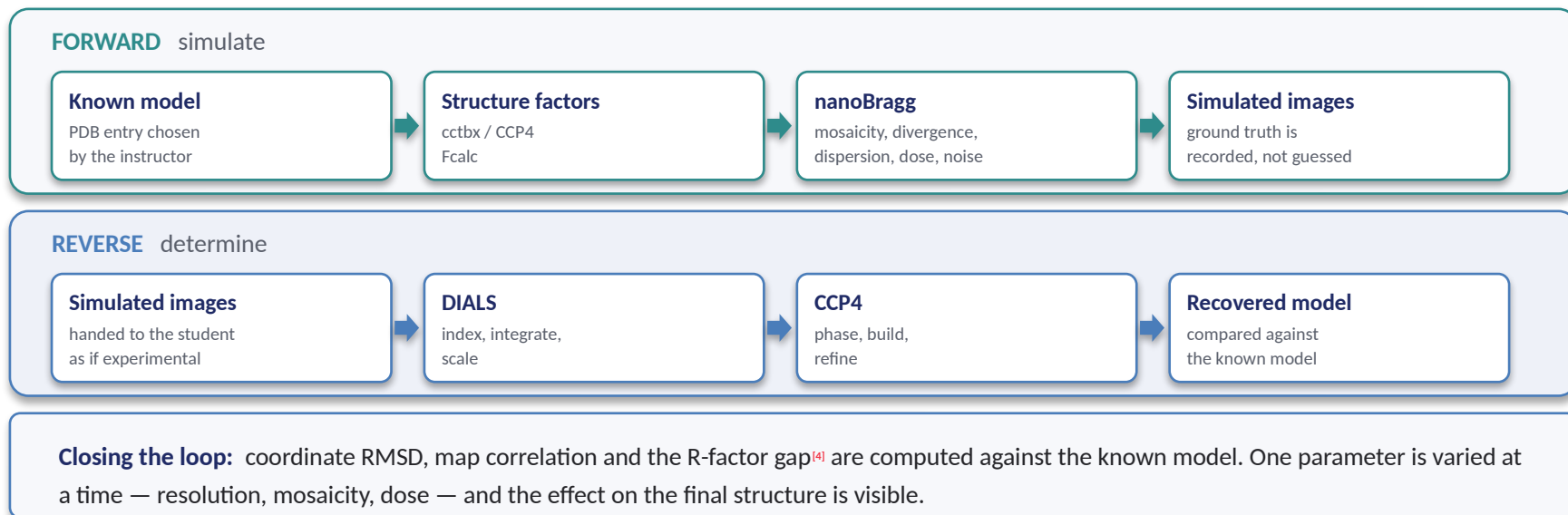
At every step the workflow surfaces the numbers, not just the files

Reads log files · evaluates $CC_{1/2}$, $I/\sigma(I)$, R_{work}/R_{free} , MolProbity scores · each figure is shown at the step that produced it, with the decision it informs

Still missing: with experimental data the student cannot check whether the answer is right. That is what the next slide solves.

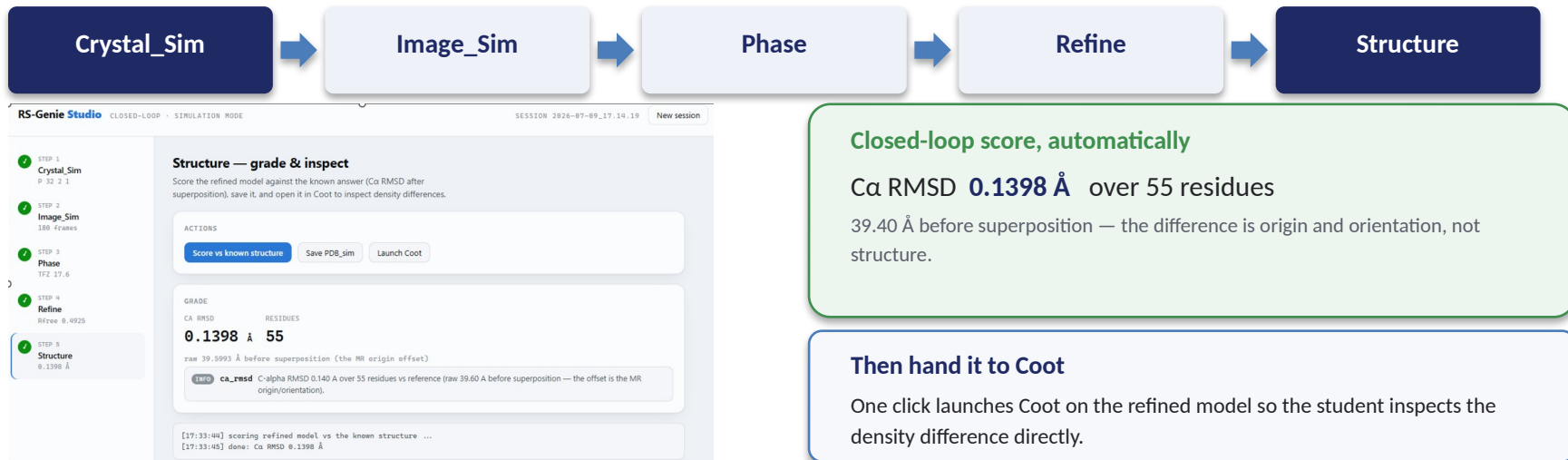
The Concrete Artefact: a nanoBragg $\rightleftharpoons$ CCP4 Closed Loop

Simulation makes the answer known in advance — the one thing real data cannot provide.



RS-Genie Studio — What Ships Today

The loop is implemented. MCP runs underneath it; the student never sees a command line and, in this version, never sees a chat box either.



Test case: brazzein R43A, a 6.4 kDa sweet protein — **PDB 7W8E**, space group $P 3_2 2 1$, 55 residues.

Kim, T.-Y., Woo, E.-J. & Yoon, T.-S. (2022). Biochem. Biophys. Res. Commun. 592, 119–124.

Tutorial Session Design — 90 Minutes, One Variable at a Time

The loop becomes a session the student drives through a guided five-step workflow.

0–15 min

Set the truth

Pick a PDB entry. Agent explains the nanoBragg parameters.

15–40 min

Generate & process

Simulate, then run DIALS and CCP4. Agent narrates each log file.

40–70 min

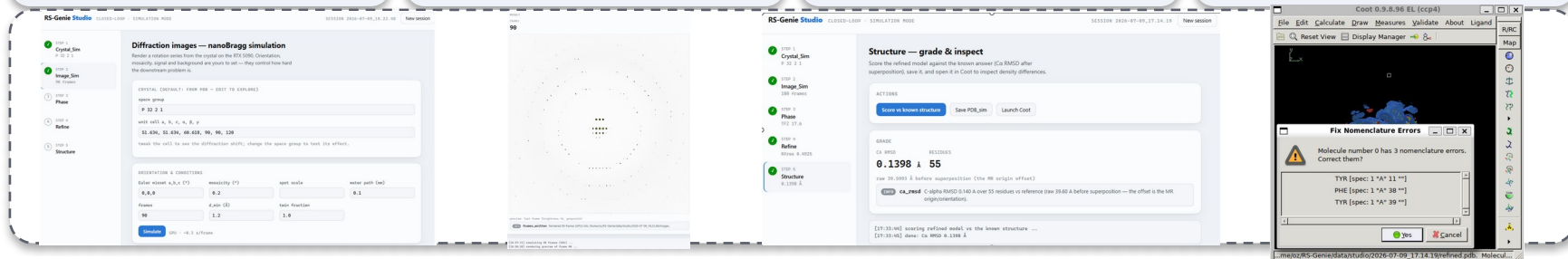
Break something

Change one parameter. Re-run. Compare against the known model.

70–90 min

Explain the gap

Why did the model degrade?
Student answers, agent checks.



No prior command-line experience is assumed at any point in the session.

Exercise set: today the loop runs on **R43A** (sweetness-diminished, P 3₂ 2 1). It extends naturally to **D29K** (sweetness-enhanced), which crystallises in P 3₁ with a different unit cell — the same small protein giving a second, genuinely different indexing problem.

Educational Impact and Classroom Use

For learners

- Follow real workflows through a guided interface
- Receive protocol-level explanation on demand
- See cause and effect against a known answer

For instructors

- Design guided exercises with reproducible state
- Inspect every command the workflow issued
- Reuse a single scaffold across cohorts

For the community

- Shared, versioned MCP service definitions
- Open contribution of new tool wrappers
- A common substrate for teaching materials

MCP as a “new genie” for crystallographic education

Supporting learners, amplifying instructors, and modernizing how crystallographic computing is taught — by giving classical crystallographic software a uniform service layer that people, and later agents, can drive.

Limits, and What Stays the Developer's Responsibility

An honest account of where this approach fails, and what must not be delegated.

Fluent but wrong

A plausible explanation of a bad decision is more dangerous than no explanation. Simulation is the mitigation: the answer is known.

Non-determinism

The same prompt can produce different tool calls. Every session logs the exact commands issued, so a result is reproducible even when the dialogue is not.

Scope of delegation

The agent selects and sequences tools. It does not implement crystallographic algorithms, and its output is never the citable result.

Does not replace teaching

It removes operational friction. Judgement about when a structure is finished still has to be taught.

Boundary: the model writes commands; the crystallographic software owns the result. Verification remains a human responsibility.

Roadmap — One Tool Layer, Two Front-Ends

The hard part is the wrapper layer. Front-ends are replaceable, which is the whole argument for building on MCP.

1

MCP service layer

DIALS, cctbx / nanoBragg and CCP4 exposed as uniform services with typed inputs and structured results.

COMPLETE

2

RS-Genie Studio

Deterministic five-step interface over those services. Closed-loop simulation, automatic scoring against the known model, Coot hand-off.

SHIPPED

3

RS-Genie

The same services driven by an LLM chat layer: free-form questions, explanation on demand, deviation from the fixed five steps.

IN DEVELOPMENT

Why this order: a deterministic front-end is reproducible and cannot hallucinate, so it is the safe thing to put in front of students first. **The same reasoning chose local weights:** a course must give the same answer next term, and a hosted model is a moving target.

Summary, and an Invitation

<https://github.com/Tae-Sung/Public>



1

The problem

Operational friction, not conceptual difficulty, is what keeps beginners out of crystallographic computing.

2

The mechanism

MCP wraps DIALS (with cctbx and nanoBragg) and CCP4 as uniform services an agent can call and explain.

3

The artefact

A nanoBragg $\rightleftharpoons$ CCP4 closed loop in which the correct answer is known, so error becomes teachable.

References and Appendix slides follow. Service definitions will be published openly through github.

If your group maintains crystallographic software, a wrapper is a small piece of work with a large teaching return. Please come and talk to me.

References

- [0] Yoon, T.-S. (2023). **Sweet Protein Crystallography in Post-AlphaFold Era: Dialogues Concerning Two ‘New’ Sciences.** Acta Cryst. A79, a179. ACA 2023 Annual Meeting, Baltimore. journals.iucr.org/a/issues/2023/a1/00/a62059/a62059.pdf
- [1] Anthropic. **Model Context Protocol (MCP) Specification.** <https://modelcontextprotocol.io>
- [2] Winter, G. et al. (2022). **DIALS as a toolkit.** Protein Science 31, 232–250.
- [3] Grosse-Kunstleve, R. W. et al. (2002). **The Computational Crystallography Toolbox: crystallographic algorithms in a reusable software framework.** J. Appl. Cryst. 35, 126–136.
- [4] Holton, J. M., Classen, S., Frankel, K. A. & Tainer, J. A. (2014). **The R-factor gap in macromolecular crystallography: an untapped potential for insights on accurate structures.** FEBS J. 281, 4046–4060.
- [5] Agirre, J. et al. (2023). **The CCP4 suite: integrative software for macromolecular crystallography.** Acta Cryst. D79, 449–461.

Note: nanoBragg is distributed within the simtbx module of cctbx — Sauter, N. K., Kern, J., Yano, J. & Holton, J. M. (2020). Acta Cryst. D76, 176–192. Phenix is outside the scope of the present implementation.

Appendix

Not presented in the talk. Kept for questions, and for readers of the published PDF who want the reasoning behind particular claims.

App. A Two design choices, one requirement

Why the model runs locally and why the loop runs in both directions — and the single requirement both answer to.

App. B Why nanoBragg, and what chemical crystallography gains

The forward direction as a beamline simulator, and why twinning and high mosaicity become teachable — with the reverse-direction software still missing.

App. C Phenix, and the Phenix AI Agent

What Phenix already ships, why their design converged on the same principles, and why a single-suite agent needs no protocol layer.

App. D How does a student catch a wrong decision?

Scoring against the known model, full command logging, and exercises in which the agent is deliberately allowed to be wrong.

App. E What hardware does this actually need?

The development machine, the deployment target, and the numeric precision that sets the memory requirement.

App. F Why build the interface before the chat?

Reproducibility, the absence of free-form command generation, and why the chat layer is cheap to add afterwards.

App. G What is inside a model file, and in what precision?

File structure, FP32 / FP16 / BF16 / FP8 and quantisation, and where integers actually occur in tokenisation.

App. H What MCP is, and how a call actually runs

The JSON-RPC client-server picture, the three primitives, one call step by step, and the $N \times M$ versus $N+M$ argument.

App. I Understanding LLMs through crystallographic language

A term-by-term correspondence with electron density, restraints, TAAM and R-free — including where the analogy breaks.

App. A — Two design choices, one requirement

It is not just a service. It has to be runnable locally and testable scientifically. Both choices below follow from that one demand.

WHY LOCAL WEIGHTS, NOT A CLOUD API

Call volume

The loop is chatty, and 'change one thing and re-run' multiplies it. Teaching is a low-value, high-frequency workload. Local inference has no marginal cost.

Access, and institutional friction

A tool needing a paid API key excludes exactly the institutions this commission most wants to reach — and sending data abroad is a procurement question, not a technical one.

A pinned dependency

A hosted model is a moving target: weights change, versions retire, and once retired a run cannot even be attempted again. Local weights are a fixed file, verifiable by digest. This pins the dependency, not the dialogue — sampling and GPU reduction order still vary. What is reproducible is the logged command sequence, and therefore the structure.

The cost we accept: a 30B local model reasons less well than a frontier one. The constrained, deterministic design is what makes that acceptable.

WHY BOTH DIRECTIONS, NOT JUST ANALYSIS

The only controlled experiment available

Real data differ in a hundred uncontrolled ways at once. Simulation is what lets a single parameter move while everything else is held fixed — which is why the loop gets re-run so often.

The forward pass teaches the physics

DIALS and CCP4 hide it behind good defaults — a student can integrate without knowing what mosaicity is. nanoBragg will not start until you have named mosaicity, divergence, dispersion and dose. You cannot simulate what you do not understand.

It makes the system testable

With no ground truth an agent cannot be evaluated at all. With simulation every run yields a measured error against the known model, so this is a test harness as much as a teaching device.

Not two decisions. One requirement — a teaching artifact must be testable, and a service cannot be.

App. B — Why nanoBragg, and what chemical crystallography can gain

The forward direction is a beamline simulator, not a spot generator — and it is the half of the loop that is not specific to proteins.

WHY nanoBragg MATTERS

Ground truth is an input, not an inference

The model, its orientation and the noise are fixed before any image exists.

The physics has to be named out loud

DIALS will integrate without you knowing mosaicity. nanoBragg will not run.

A beamline, parameter by parameter

Divergence, bandwidth, flux, exposure, detector gain and noise move alone.

One toolbox, therefore one wrapper

It ships as simtbx inside cctbx — Fcalc and images from one scriptable API.

WHAT CHEMICAL CRYSTALLOGRAPHY CAN ALSO GAIN

The forward half does not care what the atoms are

Fcalc for a small molecule or a metal cluster simulates identically.

Twinning, with the twin fraction known

Two lattices, one twin law, a chosen ratio — so BASF has a true value.

High mosaicity you can dial

Streaks, overlaps and weak spots arrive from physics, not a bad crystal.

Aspherical density, tested against the IAM

Simulate from a multipole model, refine as IAM, and measure the loss.

Why this column is not in the talk: nanoBragg takes Fcalc, so the forward half already runs for any composition. The reverse half does not — chemical crystallography closes through SHELX, olex2 and their relatives, not CCP4, and those wrappers are not written. Only the second half of the loop needs new MCP services: the N+M argument of App. H. Twinning and high mosaicity are where a known answer helps most.

nanoBragg: the simtbx module of cctbx — Sauter, N. K., Kern, J., Yano, J. & Holton, J. M. (2020). Acta Cryst. D76, 176–192.

Forward: runs today • Reverse: not yet wrapped

App. C — Phenix, and the Phenix AI Agent

Phenix is not wrapped here because the Phenix team is already building this themselves — and, independently, along the same lines.

PHENIX ALREADY SHIPS

A documentation chatbot

Answers questions over the full Phenix documentation and tutorial videos.

An 'AI Analysis' button

In the Results panel of the GUI: summarises a run and suggests next steps.

A Phenix AI Agent

Runs many Phenix programs automatically, from the GUI or the command line.

CONVERGENT DESIGN — independently

The LLM does not write commands

The model sets strategy flags; deterministic code expands them into validated parameters and strips anything outside an allowlist.

Deterministic gates between cycles

Metrics are compared against stage criteria to advance, retreat or stop.

Local execution is offered

Ollama keeps data local; the default path sends logs to a hosted provider.

Different goal, not a competing one: the Phenix agent automates production structure determination inside one suite — 23 programs already sharing one PHIL system, so it needs no protocol layer at all. RS-Genie Studio is a teaching loop that spans DIALS and CCP4, and knows the correct answer because the data were simulated. A cross-suite loop does need a protocol layer — which is where the two efforts would meet.

Source: phenix-online.org/documentation/reference/ai_agent.html · v2.2-6143, accessed 2026-08-07

A Phenix User Workshop ran at this congress, 10–11 August
— in parallel with the Quantum Crystallography workshop.

App. D — How does a student catch a wrong decision?

The answer is known

In the simulation loop the true model is on disk. Coordinate RMSD and map correlation against it are computed automatically at the end of every run.

The commands are logged

Every tool invocation is recorded with its parameters. An instructor can replay the exact sequence without the dialogue.

Disagreement is designed in

Exercises deliberately include runs where the agent's recommended resolution cut-off degrades the final model. The student is asked to find the disagreement.

The pedagogical claim: an agent that is occasionally wrong, in a setting where wrongness is detectable, teaches more than one that is always right.

App. E — What hardware does this actually need?

Two separate questions: what was used to build RS-Genie Studio, and what is needed to run it.

DEVELOPMENT — what built it

HP OMEN 45L desktop

NVIDIA RTX 5090 · 32 GB VRAM · 64 GB system RAM

Claude Code for the application code, with local models under Ollama used alongside it:

- qwen3-coder:30b 18 GB — code generation
- qwen3-vl:30b 19 GB — vision-language; the path to reading diffraction images and maps directly

Five models in total, 9.6–23 GB each. The two above carry the work; the rest were comparison runs.

DEPLOYMENT — what runs it

One workstation is enough

A 30B-class model at 4-bit is 18–23 GB and fits in 32 GB of VRAM with room for context. No server, no cluster, no external service.

Everything on a single machine

CCP4 and nanoBragg were tested under WSL on Windows 11, so one desktop hosts the model, the MCP services and the crystallographic software together. The stack ports to native Ubuntu unchanged.

Note on honesty: a hosted frontier model was used to write the software. The software itself does not require one — teaching deployments run entirely on local weights.

App. F — Why build the interface before the chat?

Reproducibility comes first

A teaching exercise has to give the same result to every student in the cohort and to the same student next week. A fixed five-step workflow guarantees that; a conversation does not.

Nothing to hallucinate

Studio issues a fixed set of parameterised calls. There is no free-form command generation, so the failure mode discussed earlier simply does not arise in the version students use today.

It makes the chat layer cheap to add

Because the tools are already MCP services, the conversational version does not reimplement anything. It replaces the front-end and inherits the same wrappers, the same logs and the same scoring.

Put plainly: MCP is the architecture, not the user interface. Students benefit from it whether or not they ever type a prompt.

App. G — What is inside a model file, and in what precision?

Two separate questions get mixed up: how the weights are stored, and how the text is represented.

FILE STRUCTURE GGUF · safetensors

Metadata header

architecture, layer count, context length, quantisation type

Tokenizer

the vocabulary itself — roughly 10^5 subword strings, plus merge rules

Tensors

the weights — essentially all of the file size

NUMERIC FORMATS of the weights

FP32 8 exp / 24 man (IEEE-754-1985: 4-byte single precision)
reference, accumulation

FP16 5 exp / 10 man (IEEE-754-2008: 2-byte half precision)
inference; narrow range

BF16 8 exp / 7 man (Google Brain team in 2017)
training + storage default

FP8 4/3 or 5/2
recent GPUs, inference

INT4 + scale ≈ 4.5 –5 effective
local deployment

30B PARAMETERS

BF16 60.0 GB

Q8 (~8.5 bit) 31.9 GB

Q4_K_M (~5 bit)

18.8 GB

pure 4-bit 15.0 GB

Observed: 18–19 GB. Q4_K_M is not pure four-bit — an FP16 scale rides on each block, and embeddings stay higher.

Tokenisation: text → subword tokens → an **integer index** into the vocabulary → that index selects one row of the embedding table, which is a **float vector**. At the output, logits over the whole vocabulary → sampling → one index → text.

The only integers are the indices. Every arithmetic operation in between is floating-point.

App. H — What MCP is, and how a call actually runs

A JSON-RPC 2.0 client-server protocol over stdio or HTTP. Often described as “USB-C for AI”: one connector instead of one adapter per pair.



THREE PRIMITIVES

- Tools** — functions the model may invoke
- Resources** — data the model may read
- Prompts** — templates the user may choose

ONE CALL, STEP BY STEP

1. **initialize** handshake, capability negotiation
2. **tools/list** server advertises what it can do
3. **tools/call** typed arguments in
4. **result** structured output back, not raw stdout

Why a protocol rather than a script: with N programs and M front-ends, bespoke glue costs $N \times M$ integrations. A shared protocol costs $N+M$. That is the whole argument — and it is why the chat layer can replace the interface without touching anything below it.

Spec: modelcontextprotocol.io · schema 2025-11-25; sessions, elicitation, sampling, streaming. Linux Foundation (AAIF) since December 2025.

App. I — Understanding LLMs through crystallographic language

Crystallographers have relied on pretrained, transferable priors since the 1960s. The vocabulary is unfamiliar; the idea is not.

In crystallography	In a language model	What is actually shared
Electron density map	The training corpus	the observations you are given
Atomic model — xyz, B	LLM model — shapes, dtypes, and weight values	a compact description that regenerates them
Scattering-factor tables	Learned representations	a small table serves every new case
Restraints — C-C = 1.52 Å	The pretrained prior	knowledge the measurement cannot supply
TAAM with MATTS — 651 atom types	Transfer to unseen inputs	a finite type library covers cases never measured
R-free on held-out reflections	Held-out cross-entropy	the guard against fitting noise
Model bias in a map	Hallucination	asserting what the data did not say

Where the analogy holds

At 2 Å the data-to-parameter ratio is only about two. Refinement works because restraint libraries — derived from small-molecule crystallography, transferred to every new structure — carry what the data cannot. Pretraining is that same distillation, at another scale.

MATTS: Jha, K. K. et al. (2022). J. Chem. Inf. Model. 62, 3752–3765.

Where it breaks

Restraints are curated and inspectable; a learned prior is neither. An $F_o - F_c$ difference map shows exactly where a model asserts what the data denies — there is no equivalent for a language model. That is why the numbers must still come from the software.

Thirty gigabytes does not contain the whole Wikipedia. It contains the contexts (regularities) that produced it —
for the same reason a coordinate file is smaller than the electron density map it explains.